



DEPARTMENT OF ELECTRIC ENERGY

TET4565 - POWER MARKET OPTIMIZATION

Project group 14

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2025-11-26

Table of Contents

List of Figures	ii
List of Tables	iii
Part 1 - Introduction	1
Formulation of the Optimization Problem	1
Data	2
Mathematical description of the optimization problem	3
Data handling	5
Part 2 - Deterministic Optimization Approaches	6
Expected Value	6
Deterministic Scenario-by-Scenario Optimization	6
Two-Stage Stochastic Optimization (Deterministic equivalent)	6
Discussion	8
Part 3 – Benders Decomposition	9
Master Problem	9
Stage 2 Subproblem (Scenario $s \in S$)	10
Benders Cuts and Algorithm	10
Results	12
Part 4 – Stochastic Dynamic Programming	13
Objectives and Learning Goals	13
Methodology	13
Single Scenario, 10 Discrete Points	14
Single Scenario, 3 Discrete Points	14
Discretization Granularity Comparison	15
Extension to All Scenarios, 10 Discrete Points	16
Comparison with Benders Decomposition	17
Comparison with Deterministic Equivalent	18
Method Selection	18
Appendix	19

List of Figures

1	Schematic diagram of the system showing wind turbines, electrolyzer, hydrogen storage, fuel cell, grid connection, and load.	1
2	Time series of wind power generation (with three scenarios), electricity price, and load demand over the 48-hour horizon.	5
3	Hydrogen tank state of charge and Fuel Cell power output.	6
4	Hourly power dispatch across wind to load, electrolyzer, and exports under the stochastic approach for the intermediate scenario.	7
5	Benders cuts plotted as linear functions of hydrogen storage H_{23} at each iteration, with the red point indicating the optimal solution at convergence.	11
6	Piecewise linear outer envelope of Stage 2 expected cost as a function of hydrogen storage H_{23} , with each colored segment representing an active Benders cut.	11
7	Stage 1 cost, Stage 2 expected cost, and total objective function as functions of hydrogen storage H_{23}	12
8	Stage 2 cost as a function of hydrogen storage for 3-point and 10-point discretizations.	15
9	Stage 1 cost, Stage 2 expected cost, and total objective function as functions of hydrogen storage across all three scenarios.	16
10	Objective function as a function of the linking variable (Benders vs SDP).	17
11	Hourly power dispatch across wind to load, electrolyzer, and exports under the Expected Value approach.	19
12	Hourly power dispatch across wind to load, electrolyzer, and exports for the intermediate scenario (deterministic).	20
13	Hourly power dispatch across wind to load, electrolyzer, and exports for the optimistic scenario (deterministic).	21
14	Hourly power dispatch across wind to load, electrolyzer, and exports for the pessimistic scenario (deterministic).	22
15	Hourly power dispatch across wind to load, electrolyzer, and exports under the stochastic approach for the optimistic scenario.	23
16	Hourly power dispatch across wind to load, electrolyzer, and exports under the stochastic approach for the pessimistic scenario.	24

List of Tables

1	Hybrid Energy System Parameters	2
2	Inflow, load and price assumptions for the 48-hour horizon	2
3	Deterministic optimized electricity costs for each scenario.	6
4	Summary of Benders iterations. Convergence achieved at iteration 7 with $ \Delta\alpha < 10^{-10}$	12
5	SDP results for single scenario with 10 discrete points.	14
6	SDP results for single scenario with 3 discrete points.	14
7	Discretization effect analysis: comparison between 10-point and 3-point SDP solutions for single scenario.	15
8	SDP results for all scenarios with 10 discrete points.	16
9	Comparison Benders vs SDP.	17
10	Comparison DE vs SDP.	18

Part 1 - Introduction

Formulation of the Optimization Problem

The system under consideration consists of a wind power plant with a fixed number of turbines. Electricity generated by the wind turbines can be directed to the load, to the electrolyzer, or exported to the grid. The system also has access to the electrical grid, which can supply energy to both the load and the electrolyzer. The load can therefore receive energy directly from the wind, from the fuel cell, or from the grid. All components are connected to a single central node.

All electricity purchased from the grid is associated with a cost, while other energy flows are assumed to be cost-free. In a short-term operational analysis, past investment expenditures are not taken into account.

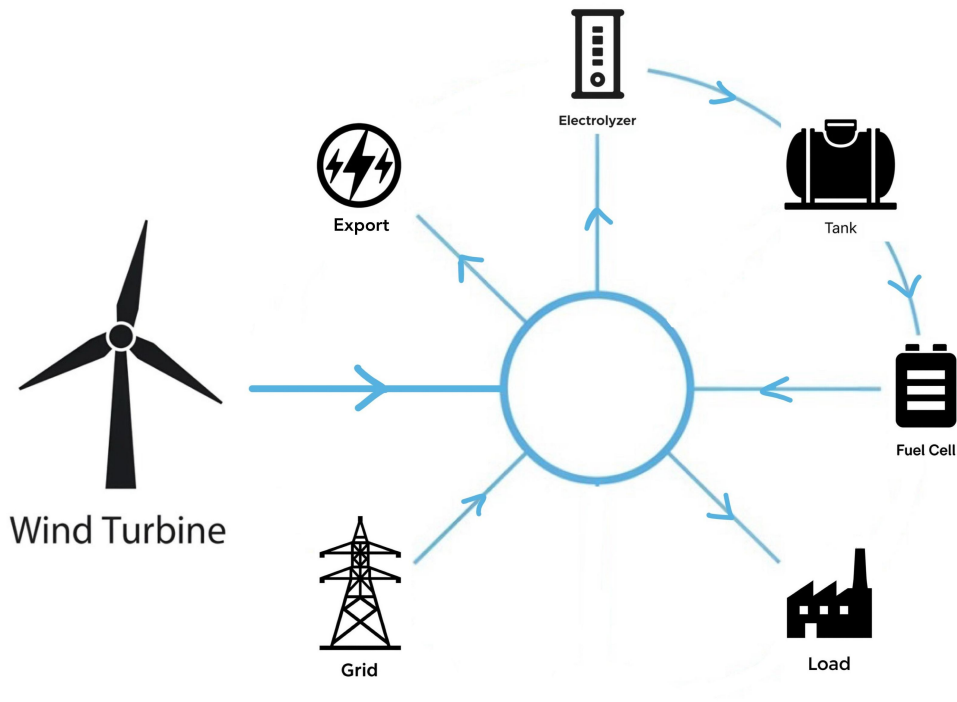


Figure 1: Schematic diagram of the system showing wind turbines, electrolyzer, hydrogen storage, fuel cell, grid connection, and load.

The system is considered over a short-term operational horizon of 48 hours, with a time step resolution of 1 hour. The objective is to minimize the cost of electricity purchased from the grid. The system data are as follows:

- **Wind Power Plant:**

The system includes a fixed number of wind turbines, resulting in a rated power determined by past investments, which is not subject to optimization. Hourly forecasts of wind power generation are available and represent the **source of uncertainty**. These forecasts are divided into Stage 1 (known) and Stage 2 (unknown), with three possible scenarios in Stage 2, each assumed equally probable.

- **Grid Export:**

Wind power can be exported to the grid when generation exceeds local demand and electrolyzer capacity. Export is limited to the available wind power and cannot be negative. The system receives a small revenue (0.01 €/kWh) for each unit of energy exported to the grid.

- **Electrolyzer:**

The electrolyzer has a fixed rated power and a given conversion efficiency, determined by prior investments. When the available power from wind and/or the grid is greater than or equal to the rated power, the electrolyzer operates at its maximum capacity. Otherwise, it operates at the available power after satisfying the load.

- **Hydrogen Tank:**

The hydrogen tank has a limited capacity due to previous investments.

- **Fuel Cell:**

The fuel cell has a fixed rated power and conversion efficiency, based on past investment decisions.

- **Load:**

The electrical demand profile is defined hourly and assumed to be known over the 48-hour horizon.

- **Electrical Grid:**

The maximum power drawn from the grid is limited to the peak load. The cost of electricity purchased from the grid is modeled as a time-dependent function:

$$C_E(t) = 50 + 4t \cdot \sin(2\pi * \frac{t}{4}) \quad [\text{€/MWh}], \quad \text{where } t \text{ is time in hours}$$

Data

Parameter	Value	Information
$P_{El,max}$	50 [kW]	Electrolyzer's rated power
η_{el}	0.7 [-]	Electrolyzer's efficiency
V_0	5 [m ³]	Starting reservoir level of the hydrogen tank
V_{MAX}	200 [kWh]	Rated capacity for the tank
$P_{FC,max}$	30 [kW]	Fuel cell's rated power
η_{FC}	0.6 [-]	Fuel cell's efficiency
C_H	1 [€/kWh]	Hydrogen value for leftover at the end of the 48th hour

Table 1: Hybrid Energy System Parameters

Parameter	Value	Information
P_W (first 24 h)	from dataset [kW]	Electric power from wind generation, given hourly.
P_W^s (last 24 h)	from dataset [kW]	Electric power from wind generation, given hourly. This includes uncertainty.
$N_{scenarios}$	3	Number of scenarios for the second stage (last 24 hours).
$p_{scenario}$	1/3	The probability for each scenario.
C_E	$50 + 4t \cdot \sin(2\pi * \frac{t}{4})$ [€/MWh]	Price of energy bought from the grid. Power price for all 48 hours.
P_{Load} (48 h)	from dataset [kW]	Load demand given hourly.
$C_{Exp,i,j}$	0.01 [€/kWh]	Revenue from energy exported to the grid.

Table 2: Inflow, load and price assumptions for the 48-hour horizon

Mathematical description of the optimization problem

Sets and Parameters

- $T_i = \{0,1,2,\dots,23\}$: time interval representing first stage (first 24 hours)
- $T_j = \{24,25,26,\dots,47\}$: time interval representing second stage (second 24 hours)
- $\mathbf{t} = T_i \cup T_j = \{0,1,2,\dots,47\}$: time index (total 48 hours)
- $s = 1, 2, 3$: scenario index for wind generation
- $P_w^s(t)$: power generated by wind turbines at time t under scenario s [MW]
- p_s : probability of scenario s , with $\sum_{s=1}^3 p_s = 1$
- $P_{\text{El,max}}$: rated power of the electrolyzer [MW]
- η_{el} : efficiency of the electrolyzer (conversion to hydrogen)
- H_{max} : maximum storage capacity of the hydrogen tank [kg]
- $L(t)$: electricity demand of the load at time t [MW]
- $C_E(t)$: electricity price from the grid at time t [€/MWh]
- $P_{\text{FC,max}}$: maximum power output of the fuel cell [MW]
- η_{fc} : efficiency of the fuel cell (conversion from hydrogen to electricity)
- Δt : time step duration [h]
- $H^s(t)$: hydrogen stored in the tank at time t under scenario s [kg]
- $H_{\text{use}}^s(t)$: hydrogen used by the fuel cell at time t under scenario s [kg]
- $P_{\text{W.to.Load}}^s(t)$: wind power directly supplied to the load at time t under scenario s [MW]

Decision Variables

- $P_{\text{El}}^s(t)$: electrolyzer power at time t under scenario s [MW]
- $P_{\text{FC}}^s(t)$: fuel cell power supplied to the load at time t under scenario s [MW]
- $P_{\text{Grid}}^s(t)$: electricity purchased from the grid at time t under scenario s [MW]
- $P_{\text{Export}}^s(t)$: Exported wind power at time t under scenario s [MW]

Constraints

- Export

$$P_{\text{Export}}^s(t) \leq \max(P_W^s) \quad \forall t, s \quad (1)$$

$$P_{\text{Export}}^s(t) \geq 0 \quad \forall t, s \quad (2)$$

- Electrolyzer Operation

$$P_{\text{El}}^s(t) \leq P_{\text{El},\max} \quad \forall t, s \quad (3)$$

$$P_{\text{El}}^s(t) \geq 0 \quad \forall t, s \quad (4)$$

- Hydrogen Storage Dynamics

$$H^s(t) \leq H_{\max} \quad \forall t, s \quad (5)$$

$$H^s(t) \geq 0 \quad \forall t, s \quad (6)$$

$$H^s(t+1) = H^s(t) + \eta_{\text{El}} P_{\text{El}}^s(t+1) \Delta t - \frac{P_{\text{FC}}^s(t+1) \cdot \Delta t}{\eta_{\text{FC}}} \quad \forall t, s \quad (7)$$

- Fuel Cell Constraints

$$P_{\text{FC}}^s(t) \leq P_{\text{FC},\max} \quad \forall t, s \quad (8)$$

$$P_{\text{FC}}^s(t) \geq 0 \quad \forall t, s \quad (9)$$

- Grid Supply

$$P_{\text{Grid}}^s(t) \leq \max(P_{\text{Load}}) \quad \forall t, s \quad (10)$$

$$P_{\text{Grid}}^s(t) \geq 0 \quad \forall t, s \quad (11)$$

- Load Balance

$$P_{\text{Load}}(t) = P_W^s(t) - P_{\text{Export}}^s(t) - P_{\text{El}}^s(t) + P_{\text{Grid}}^s(t) + P_{\text{FC}}^s(t) \quad \forall t, s \quad (12)$$

Expressions

- Wind Balance

$$P_W^s(t) = P_{W \rightarrow \text{Load}}^s(t) + P_{\text{El}}^s(t) + P_{\text{Export}}^s(t)$$

Objective Function

Minimize the expected total cost of electricity purchased from the grid:

$$\min \sum_{i=0}^{23} (C_{E,i} P_{\text{Grid},i} - C_{\text{Exp},i} P_{\text{Exp},i}) + \sum_{s=1}^3 p_s \left[\sum_{j=24}^{47} (C_{E,j} P_{\text{Grid},j}^s - C_{\text{Exp},j} P_{\text{Exp},j}^s) - C_H H_{47} \right]$$

Each scenario s represents a possible realization of wind generation over the 48 hours. Probabilities p_s are equal for each scenario.

The stochastic optimization balances the use of hydrogen versus purchasing electricity from the grid under uncertainty.

Linking variable

$H^s(t)$ is the linking variable so when the stage 1 ends the stage 2 begins and that variable is transmitted:

$$H^s(j) = H^s(i) + \eta_{\text{El}} P_{\text{El}}^s(j) \Delta t - \frac{P_{\text{FC}}^s(j) \cdot \Delta t}{\eta_{\text{FC}}} \quad \text{for } i = 23 \text{ and } j = 24, \forall s$$

Data handling

It was written a Python code that performs data extraction, processing, and visualization for wind power generation, electricity consumption, and theoretical price over a 48-hour period.

The script first imports a CSV file containing time-series data, using the first column as a datetime index to enable precise temporal selection. Wind generation data are extracted for two distinct days—January 2 and January 3, 2020—creating individual daily series. For the second day, two modified scenarios are generated by scaling the original wind values by factors of 1.2 (optimistic scenario) and 0.7 (pessimistic scenario), representing potential variations in wind availability.

Electricity consumption data are also extracted from the same dataset, covering both days as a continuous load series. A time vector with hourly resolution is generated to align all series over the 48-hour horizon. Theoretical electricity prices are computed as a function of time to simulate a dynamic pricing profile.

Finally, the code produces a figure comprising three vertically stacked subplots with a shared x-axis. The first subplot displays wind production for both days and the associated scenarios. The second subplot presents the theoretical price evolution over the same period. The third subplot illustrates the electricity load profile. All plots include grid lines, labels, and legends to facilitate interpretation. The procedure allows a clear visual comparison of wind generation variability, load demand, and price signals over the 48-hour period.

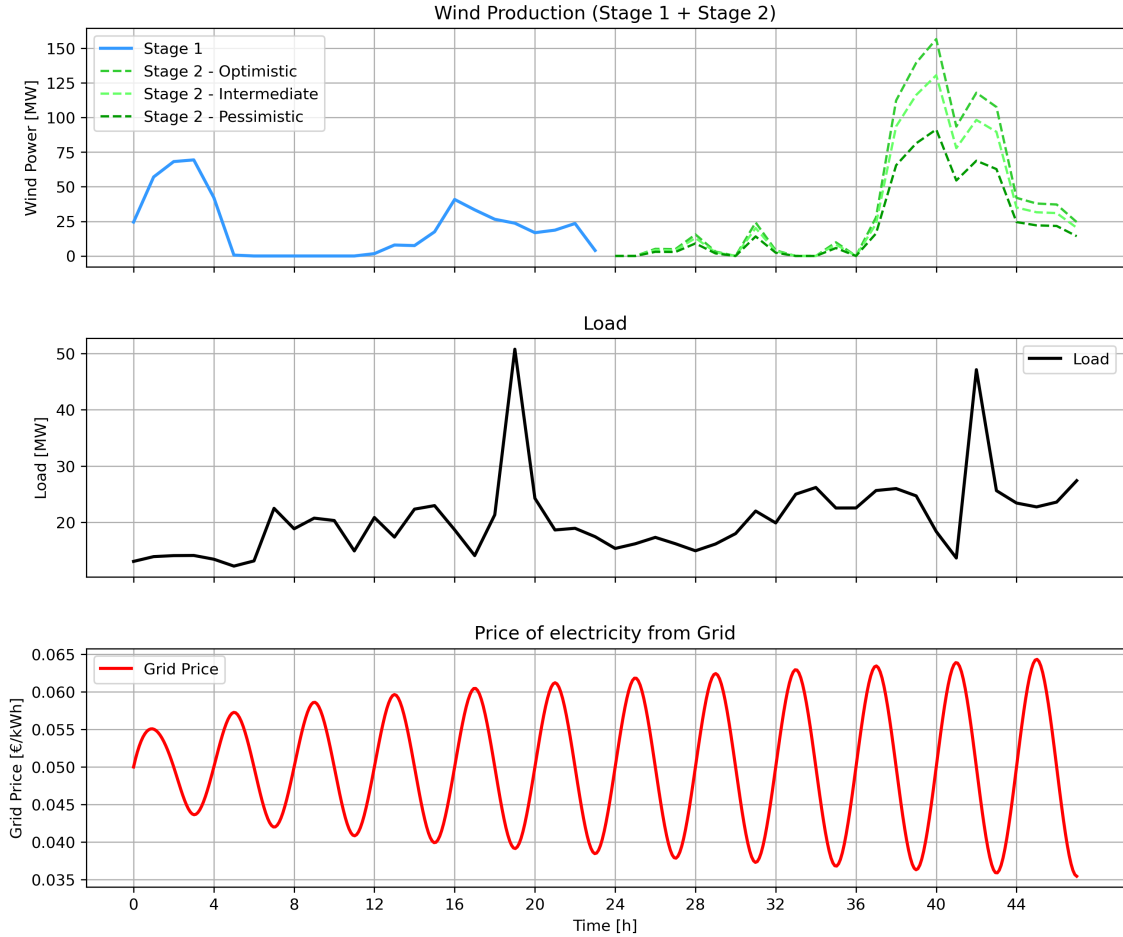


Figure 2: Time series of wind power generation (with three scenarios), electricity price, and load demand over the 48-hour horizon.

Part 2 - Deterministic Optimization Approaches

Expected Value

The first variant employs expected value optimization, computing the solution by averaging wind generation values across three scenarios (intermediate, optimistic, pessimistic) for each hour. The optimization problem is formulated deterministically using these aggregated values.

This method yielded a total electricity cost of **15.6846 €**.

Deterministic Scenario-by-Scenario

The second variant solves the optimization problem independently for each scenario, treating each as a distinct deterministic case with unit probability. This approach permits the model to fully adapt to the specific wind realization within each scenario, resulting in scenario-dependent solutions. The optimized electricity costs for each scenario are presented below.

Scenario	Optimized Cost
Pessimistic	16.7483 €
Intermediate	15.5557 €
Optimistic	14.9673 €

Table 3: Deterministic optimized electricity costs for each scenario.

Two-Stage Stochastic Optimization (Deterministic equivalent)

The stochastic formulation implements a 48-hour optimization problem with a “here-and-now” decision structure, wherein each scenario contributes to the objective function proportionally to its probability. Under this framework, the fuel cell behavior is determined uniquely for the first day, whereas the hydrogen tank state of charge varies across the three scenarios in the second day, with each scenario’s outcome weighted equally in the final objective calculation. This approach minimizes the expected total electricity cost:

$$\text{obj} = \text{cost}_{\text{day1}} + \text{cost}_{\text{day2}} = \mathbf{8.7921 \text{ €}} + \mathbf{6.9818 \text{ €}} = \mathbf{15.7738 \text{ €}}$$

The figure below illustrates the fuel cell power output decisions and the corresponding hydrogen tank state of charge evolution, displaying a single trajectory for the first day and three branching trajectories for the second day across the respective scenarios.

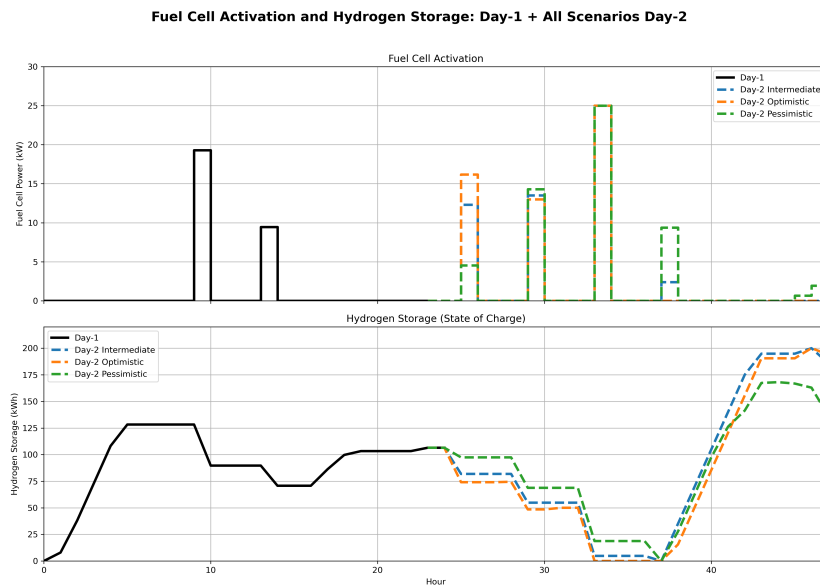


Figure 3: Hydrogen tank state of charge and Fuel Cell power output.

Figure 4 illustrates the primary allocation variables over 48 hours resulting from the here-and-now decision structure. This dispatch represents the solution obtained through the deterministic equivalent of the two-stage stochastic problem, constituting the unique resolution to the stochastic optimization framework. While all scenario-specific dispatches are provided in the appendix, only this comprehensive dispatch is presented in the main text for clarity. It is important to note that displaying the dispatch across the full 48-hour horizon requires the realization of a specific scenario; the intermediate scenario is selected for this illustration. Consequently, this dispatch does not represent the stochastic problem solution itself, but rather clarifies the temporal evolution of all optimization variables under a particular scenario realization.

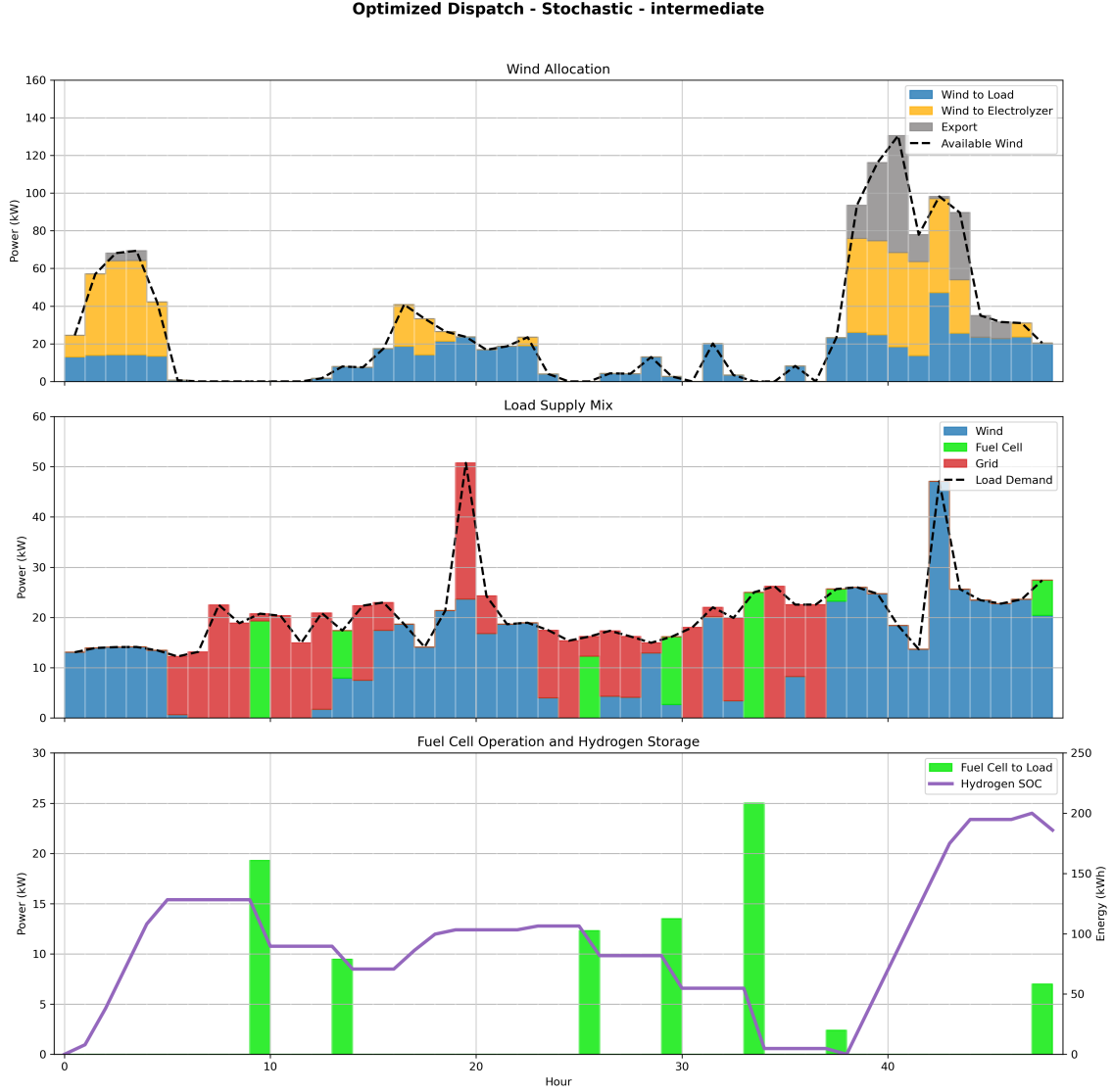


Figure 4: Hourly power dispatch across wind to load, electrolyzer, and exports under the stochastic approach for the intermediate scenario.

Discussion

Comparison between Expected Value and Individual Scenarios

The Expected Value (EV) approach delivers a total cost of **15.68 €**, which is slightly higher than the intermediate scenario (**15.56 €**). This difference is reasonable because the three scenarios are asymmetric: the optimistic case scales wind by 1.2, while the pessimistic case scales it by 0.7. As a result, the average wind profile is less favorable than the intermediate scenario.

Comparison between Deterministic Scenarios

When each scenario is optimized independently, the model fully adapts to the specific wind realization. Consequently, the optimistic scenario achieves the lowest cost (**14.97 €**), while the pessimistic scenario reaches the highest cost (**16.75 €**). The intermediate scenario lies between these extremes at **15.56 €**. These results confirm that deterministic optimization is highly sensitive to wind availability: more wind reduces grid purchases and overall cost, while less wind increases reliance on the grid.

Comparison between Expected Value and Stochastic Approach

The two-stage stochastic approach yields a total cost of **15.77 €**, marginally higher than the EV solution (**15.68 €**). This seemingly counterintuitive result reflects a fundamental principle in stochastic optimization: enforcing non-anticipativity has a cost.

The Expected Value approach optimizes Stage 1 decisions using averaged scenario data, then applies optimal Stage 2 decisions for each individual scenario independently. In contrast, the stochastic formulation requires Stage 1 decisions to remain feasible under all three scenarios simultaneously.

Specifically, the pessimistic scenario ($s = 3$: low wind generation) necessitates higher grid purchases in Stage 1. This forces a more conservative first-stage strategy to guarantee robustness, increasing overall operational cost. The difference ($15.77 \text{ €} - 15.68 \text{ €} = 0.09 \text{ €}$) represents the price of robustness: the premium paid to eliminate recourse violations and ensure implementation feasibility.

Validation of Stochastic Formulation

To validate the stochastic model, its deterministic equivalent has been tested by running the optimization for each single scenario and comparing the results with the corresponding deterministic subtasks (that means running the optimization setting the probability 1 to one scenario and 0 to the others two). The costs matched exactly, confirming that the stochastic formulation correctly reproduces scenario-specific outcomes when probabilities collapse to one. This consistency demonstrates that the model is well-structured and that differences in cost between deterministic and stochastic approaches arise solely from the non-anticipativity constraint and uncertainty hedging.

Key Insight

Overall, the comparison highlights a fundamental trade-off: deterministic solutions minimize cost for a given scenario but ignore uncertainty, while the stochastic approach sacrifices a small amount of optimality in the first day to ensure robustness across all possible realizations. This trade-off is crucial in energy systems planning, where variability in renewable generation can significantly impact operational feasibility and cost.

Part 3 – Benders Decomposition

Master Problem

The master problem optimizes first-stage decisions over hours $t \in T_i = \{0, \dots, 23\}$, including wind-to-electrolyzer allocation, fuel cell dispatch, grid purchases, and hydrogen storage. It minimizes the sum of Stage 1 grid costs and the auxiliary variable α , which represents the expected Stage 2 cost.

Decision variables:

$P_{El}(t)$	Power from wind to electrolyzer at time $t \in T_i$
$P_{FC}(t)$	Power from fuel cell to load at time $t \in T_i$
$P_{Grid}(t)$	Power from the grid at time $t \in T_i$
$P_{Export}(t)$	Export wind power at time $t \in T_i$

Objective function:

$$\begin{aligned} \min_{P, H, \alpha} \quad & \sum_{i=0}^{23} (C_{E,i} P_{Grid,i} - C_{Exp,i} P_{Export,i}) + \alpha \\ \text{s.t.} \quad & 0 \leq P_{Export}(t) \leq \max(P_W(t)) \quad \forall t \in T_i, \\ & 0 \leq P_{El}(t) \leq P_{El,max} \quad \forall t \in T_i, \\ & H_0 = 0 \\ & H(t+1) = H(t) + \eta_{El} P_{El}(t) \Delta t - \frac{P_{FC}(t) \Delta t}{\eta_{FC}} \quad \forall t \in T_i, \\ & 0 \leq H(t) \leq H_{max} \quad \forall t \in T_i, \\ & 0 \leq P_{FC}(t) \leq P_{FC,max} \quad \forall t \in T_i, \\ & 0 \leq P_{Grid}(t) \leq P_{Grid,max} \quad \forall t \in T_i, \\ & P_{Load}(t) = P_W(t) - P_{Export}(t) - P_{El}(t) + P_{Grid}(t) + P_{FC}(t) \quad \forall t \in T_i. \end{aligned}$$

Thus, H_{23} is obtained as a result of the first-stage optimization and is used as the initial hydrogen level for the Stage 2 subproblems.

Stage 2 Subproblem (Scenario $s \in S$)

Each subproblem optimizes second-stage decisions over hours $t \in T_j = \{24, \dots, 47\}$, given the hydrogen level H_{23} from Stage 1.

Decision variables:

$P_{\text{El}}^s(t)$	Power from wind to electrolyzer at time $t \in T_j$
$P_{\text{FC}}^s(t)$	Power from fuel cell to load at time $t \in T_j$
$P_{\text{Grid}}^s(t)$	Power from the grid at time $t \in T_j$
$P_{\text{Export}}^s(t)$	Export wind power at time $t \in T_j$

Objective function:

$$\begin{aligned}
& \min_{P, H} \sum_{t \in T_j} (C_{E,t} P_{\text{Grid},t}^s - C_{\text{Exp},t} P_{\text{Export},t}^s) - C_H H_{47}^s \\
& \text{s.t. } H^s(24) = H_{23} + \eta_{\text{El}} P_{\text{El}}^s(24) \Delta t - \frac{P_{\text{FC}}^s(24) \Delta t}{\eta_{\text{FC}}}, \\
& \quad H^s(t+1) = H^s(t) + \eta_{\text{El}} P_{\text{El}}^s(t+1) \Delta t - \frac{P_{\text{FC}}^s(t+1) \Delta t}{\eta_{\text{FC}}} \quad \forall t \in T_j, \\
& \quad 0 \leq H^s(t) \leq H_{\max} \quad \forall t \in T_j, \\
& \quad 0 \leq P_{\text{El}}^s(t) \leq P_{\text{El},\max} \quad \forall t \in T_j, \\
& \quad 0 \leq P_{\text{FC}}^s(t) \leq P_{\text{FC},\max} \quad \forall t \in T_j, \\
& \quad 0 \leq P_{\text{Grid}}^s(t) \leq P_{\text{Grid},\max} \quad \forall t \in T_j, \\
& \quad 0 \leq P_{\text{Export}}^s(t) \leq P_{\text{W}}^s(t) \quad \forall t \in T_j, \\
& \quad P_{\text{Load}}^s(t) = P_{\text{FC}}^s(t) + P_{\text{Grid}}^s(t) + P_{\text{W}}^s(t) - P_{\text{Export}}^s(t) - P_{\text{El}}^s(t) \quad \forall t \in T_j.
\end{aligned}$$

The optimal cost of each subproblem is used to compute:

$$\phi(H_{23}) = \sum_{s \in S} p_s \cdot \text{OptimalCost}_{\text{Stage2}}^s$$

Benders Cuts and Algorithm

The Benders decomposition method constructs an iterative lower bound approximation of the expected Stage 2 cost function through the addition of optimality cuts. At each iteration k , after solving all Stage 2 subproblems for a given hydrogen level $\tilde{H}_{23}^k$ obtained from the master problem, a new cut is generated and added to the master:

$$\alpha \geq \varphi^k + \lambda^k \left(H_{23} - \tilde{H}_{23}^k \right),$$

where $\varphi^k = \sum_{s \in S} p_s \cdot Q_s(\tilde{H}_{23}^k)$ is the expected Stage 2 cost evaluated at the current hydrogen level, and $\lambda^k = \sum_{s \in S} p_s \cdot \lambda_s^k$ is the probability-weighted dual multiplier of the linking constraint across all scenarios. The auxiliary variable α in the master problem is automatically computed as the maximum over all existing cuts, providing an increasingly accurate lower bound on the true expected Stage 2 cost.

The algorithm proceeds iteratively:

1. **Solve the master problem** to obtain Stage 1 decisions and hydrogen level H_{23} .
2. **Solve each Stage 2 subproblem** for all scenarios, using H_{23} as the initial hydrogen state. Compute the expected cost $\varphi(H_{23})$ and extract dual multipliers λ_s .
3. **Generate and add a Benders cut** to the master problem.
4. **Check convergence:** if $|\alpha - \varphi(H_{23})| \leq \epsilon$ (with $\epsilon = 10^{-10}$ €), terminate; otherwise, return to step 1.

To visualize the Benders cuts, Figure 5 displays all cuts as linear functions of H_{23} at each iteration, with the red point representing the final optimal solution.

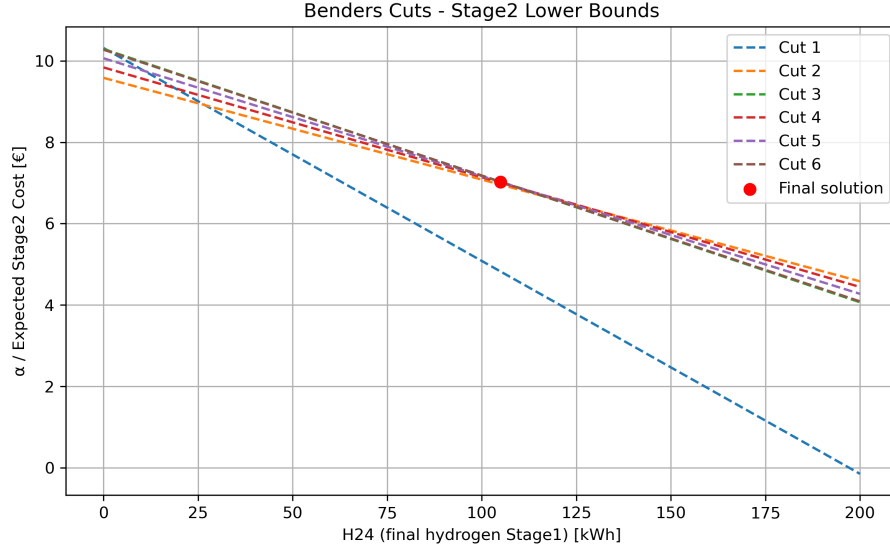


Figure 5: Benders cuts plotted as linear functions of hydrogen storage H_{23} at each iteration, with the red point indicating the optimal solution at convergence.

The piecewise linear approximation is further illustrated in Figure 6, which shows the outer envelope of all cuts, with each colored segment representing an active constraint.

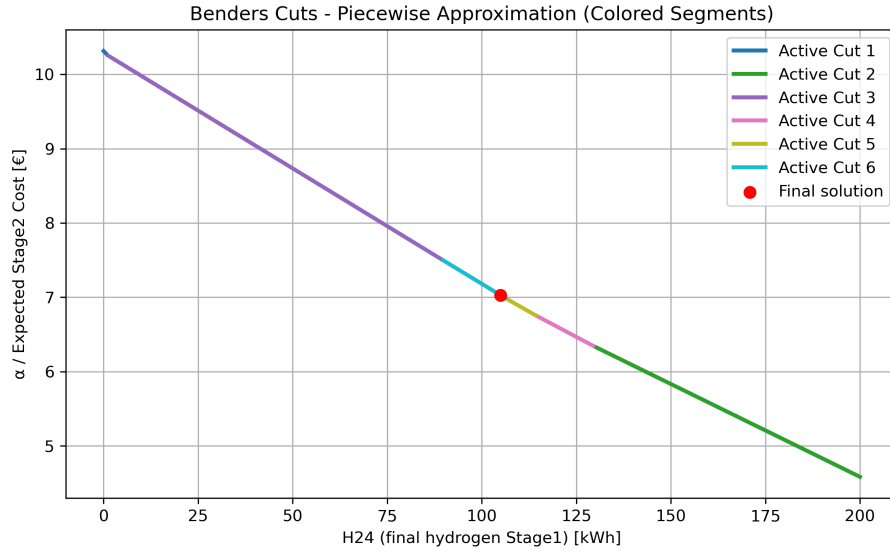


Figure 6: Piecewise linear outer envelope of Stage 2 expected cost as a function of hydrogen storage H_{23} , with each colored segment representing an active Benders cut.

Results

The Benders decomposition algorithm converges in seven iterations, with Table 4 documenting the progression of key variables throughout the process.

Iter.	H_{23} [kWh]	α [€]	E[Stage 2] [€]	$ \Delta\alpha $	λ [€/kWh]	Gap [€]
1	0.00	0.000	10.315	1.031e+01	-0.0523	10.315
2	163.97	1.735	5.482	3.747e+00	-0.0250	3.747
3	80.36	7.572	7.790	2.178e-01	-0.0311	0.218
4	115.96	6.682	6.710	2.728e-02	-0.0270	0.028
5	109.38	6.887	6.897	9.602e-03	-0.0290	0.010
6	104.95	7.025	7.028	2.959e-03	-0.0309	0.003
7	106.45	6.982	6.982	7.105e-15	—	0.000

Table 4: Summary of Benders iterations. Convergence achieved at iteration 7 with $|\Delta\alpha| < 10^{-10}$.

Throughout the iterative process, all dual multipliers λ^k were observed to be negative. This behavior is economically meaningful: a negative dual indicates that increasing the hydrogen level H_{23} reduces the expected Stage 2 cost. The magnitude of the dual multiplier reflects the marginal economic value of hydrogen storage. The progression from -0.0523 to -0.0309 demonstrates diminishing marginal returns as additional hydrogen provides less benefit at higher storage levels due to saturation effects and fuel cell capacity constraints.

The optimal hydrogen level at the end of Stage 1 is: $H_{23}^* = 106.45$ kWh

This leads to a minimum value of the objective function of:

$$\text{Obj} = \text{Stage 1 cost} + \text{Stage 2 expected cost} = 8.7921 \text{ €} + 6.9818 \text{ €} = \mathbf{15.7738 \text{ €}}$$

The required computational time was **0.50 seconds**.

Figure 7 visualizes how the Stage 1 cost, Stage 2 expected cost, and total objective function evolve as functions of the hydrogen linking variable, revealing the characteristic structure of the problem landscape.

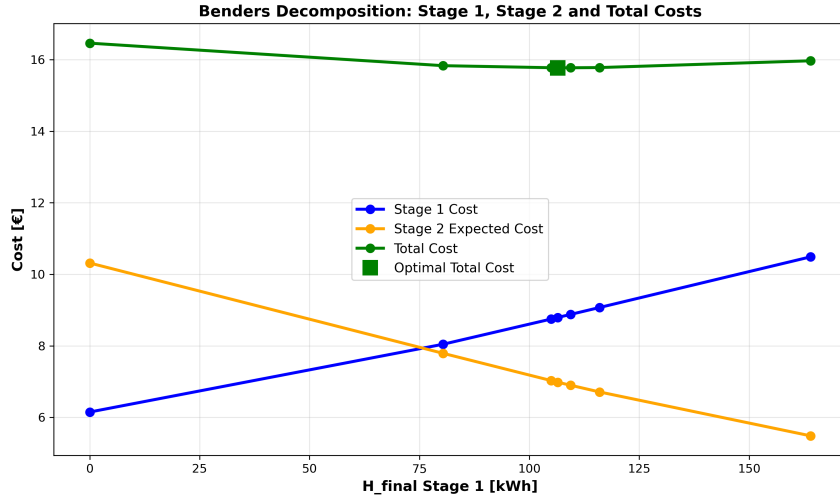


Figure 7: Stage 1 cost, Stage 2 expected cost, and total objective function as functions of hydrogen storage H_{23} .

The optimal dispatch obtained via Benders decomposition is identical to the two-stage stochastic optimization results from Part 2, confirming the correctness and internal consistency of the decomposition approach. This agreement validates both the Benders algorithm implementation and the overall modeling framework.

Part 4 – Stochastic Dynamic Programming

Objectives and Learning Goals

This section tackles the two-stage hydrogen storage optimization problem using Stochastic Dynamic Programming (SDP), investigating the interplay between discretization granularity, scenario aggregation, and computational efficiency. The core objectives are:

- Understand the fundamental differences between SDP and Benders decomposition as decomposition techniques for multi-stage stochastic optimization;
- Implement SDP by explicitly constructing value functions through discretization of the linking variable (hydrogen storage at the end of Stage 1);
- Quantify the impact of discretization levels (3 vs. 10 points) on solution accuracy and computational cost;
- Assess the robustness gains from scenario aggregation (single vs. all three scenarios);
- Validate SDP results against those from Benders decomposition and stochastic two-stage optimization.

Methodology

The SDP approach treats the linking variable—hydrogen storage at hour 23 (H_{23})—as the state variable connecting Stage 1 and Stage 2. The key steps are:

1. ****State Space Discretization:**** The continuous range $[0, H_{\max}]$ is divided into N equally spaced discrete points. We explore $N = 3$ (coarse) and $N = 10$ (fine) discretizations.
2. ****Stage 1 Optimization:**** For each candidate value $H_{23}^{(i)}$ (where $i \in \{1, 2, \dots, N\}$), the first-day problem (hours 0–23) is solved with a hard constraint:

$$H_t(i)|_{t=23} = H_{23}^{(i)}.$$

This yields the Stage 1 cost $C_1^{(i)}$.

3. ****Stage 2 Optimization:**** Given $H_{23}^{(i)}$ as initial condition, the second-day problem (hours 24–47) is solved under each scenario $s \in \{\text{intermediate, optimistic, pessimistic}\}$, yielding scenario-specific Stage 2 costs $C_2^{(i,s)}$.
4. ****Expected Cost Aggregation:**** The expected Stage 2 cost for each state is:

$$\mathbb{E}[C_2^{(i)}] = \sum_s p_s \cdot C_2^{(i,s)},$$

where $p_s = 1/3$ for each scenario (equal probability assumption).

5. ****Total Cost and Optimality:**** The total cost for each state is:

$$\text{Total}^{(i)} = C_1^{(i)} + \mathbb{E}[C_2^{(i)}].$$

The optimal linking variable is:

$$H_{23}^* = \arg \min_i \text{Total}^{(i)}.$$

This approach explicitly constructs the value function $V(H_{23})$ by connecting the discrete points. It provides full transparency on the trade-off between Stage 1 and Stage 2 costs across different hydrogen inventory levels, offering intuitive insights into the optimal balancing of present and future decisions under uncertainty.

Single Scenario, 10 Discrete Points

The problem is solved for a single stochastic scenario (intermediate wind profile) with **10** equally spaced discrete points spanning the hydrogen storage range $[0, 200]$ kWh.

Results Table

State	H_{23} [kWh]	Stage 1 Cost [€]	Stage 2 Cost [€]	Total Cost [€]
1	0.00	6.148	10.107	16.255
2	22.22	6.590	9.407	15.996
3	44.44	7.145	8.707	15.852
4	66.67	7.701	8.011	15.712
5	88.89	8.287	7.319	15.607
6	111.11	8.928	6.632	15.561
7	133.33	9.580	6.058	15.638
8	155.56	10.237	5.502	15.739
9	177.78	11.247	4.947	16.193
10	200.00	12.477	4.391	16.868

Table 5: SDP results for single scenario with 10 discrete points.

Optimal Solution

The optimal hydrogen level at the end of Stage 1 is: $H_{23}^* = 111.11$ kWh

That leads to a minimum value of the objective function of:

$$\text{Obj} = \text{Stage 1 cost} + \text{Stage 2 cost} = 8.928 \text{ €} + 6.632 \text{ €} = \mathbf{15.561 \text{ €}}$$

To reach this result, the required computational time was **0.60 seconds**.

Single Scenario, 3 Discrete Points

To investigate the effect of discretization coarseness, the same problem is solved with only **3** equally spaced discrete points: empty, half-full, and full storage.

Results Table

State	H_{23} [kWh]	Stage 1 Cost [€]	Stage 2 Cost [€]	Total Cost [€]
1	0.00	6.148	10.107	16.255
2	100.00	8.606	6.976	15.581
3	200.00	12.477	4.391	16.868

Table 6: SDP results for single scenario with 3 discrete points.

Optimal Solution

The optimal hydrogen level at the end of Stage 1 is: $H_{23}^* = 100.00$ kWh

That leads to a minimum value of the objective function of:

$$\text{Obj} = \text{Stage 1 cost} + \text{Stage 2 cost} = 8.606 \text{ €} + 6.976 \text{ €} = \mathbf{15.581 \text{ €}}$$

To reach this result, the required computational time was **0.15 seconds**.

Discretization Granularity Comparison

Figure 8 visualizes the Stage 2 cost functions as learned by both discretizations. By connecting the discrete (H_{23}, C_2) points, we construct piecewise-linear approximations of the underlying value function.

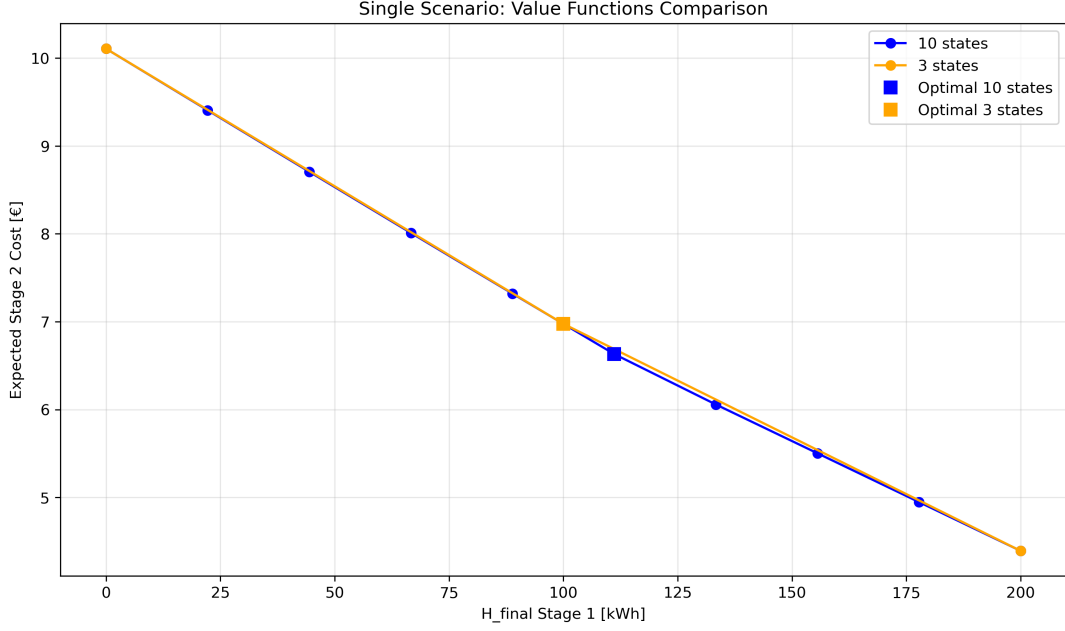


Figure 8: Stage 2 cost as a function of hydrogen storage for 3-point and 10-point discretizations.

Comparing the 3-point and 10-point solutions:

Metric	10 Points	3 Points	Relative Change
Optimal H_{23} [kWh]	111.11	100.00	-9.99%
Total Cost [€]	15.561	15.581	+0.013%
Computational Time [s]	0.60	0.15	-75%

Table 7: Discretization effect analysis: comparison between 10-point and 3-point SDP solutions for single scenario.

The 10-point discretization produces a smoother curve, while the 3-point version captures the main structure with fewer evaluations. Although the optimal hydrogen levels differ (111.11 vs 100.00 kWh), the total costs are nearly identical, differing by only 0.013%. This indicates that the objective function is relatively flat in the optimal region so that small shifts in the linking variable do not significantly affect the final cost.

The 3-point solution is 75% faster, which demonstrates the computational trade-off. With negligible cost loss and significant speed-up, a coarser discretization is practical for quick assessments. However, the 10-point approach remains very efficient at 0.60 seconds and provides better accuracy with minimal computational overhead.

For this hydrogen storage problem, both discretizations lead to essentially equivalent solutions, which reflects its relatively simple structure. The cost function exhibits a clear, broad minimum without sharp fluctuations or multiple local optima. Based on this observation, we hypothesize that discretization granularity may be less critical in simple, well-behaved problems; however, more complex systems with multiple constraints or nonlinear dynamics might require finer discretization to avoid missing important features of the value function. For the present case, the 10-point discretization offers a practical balance, while the 3-point case is useful for rapid feasibility checks.

Extension to All Scenarios, 10 Discrete Points

To assess robustness under uncertainty, SDP is extended to incorporate all three wind scenarios (intermediate, optimistic, pessimistic) with equal probabilities $p_s = 1/3$. The Stage 2 cost for each hydrogen state becomes the probability-weighted average of scenario-specific costs.

State	H_{23} [kWh]	Stage 1 Cost [€]	Stage 2 Cost [€]	Total Cost [€]
1	0.00	6.148	10.315	16.463
2	22.22	6.590	9.614	16.203
3	44.44	7.145	8.914	16.059
4	66.67	7.701	8.217	15.918
5	88.89	8.287	7.526	15.813
6	111.11	8.928	6.847	15.776
7	133.33	9.580	6.248	15.828
8	155.56	10.237	5.693	15.929
9	177.78	11.247	5.137	16.384
10	200.00	12.477	4.581	17.059

Table 8: SDP results for all scenarios with 10 discrete points.

The optimal hydrogen level at the end of Stage 1 is: $H_{23}^* = 111.11$ kWh

That leads to a minimum value of the objective function of:

$$\text{Obj} = \text{Stage 1 cost} + \text{Stage 2 expected cost} = 8.928 \text{ €} + 6.847 \text{ €} = \mathbf{15.776 \text{ €}}$$

To reach this result, the required computational time was **1.25 seconds**.

Figure 9 visualizes how Stage 1 cost, Stage 2 expected cost, and the total objective function vary across the discretized hydrogen levels, revealing the characteristic U-shaped curve with optimality at state 6.

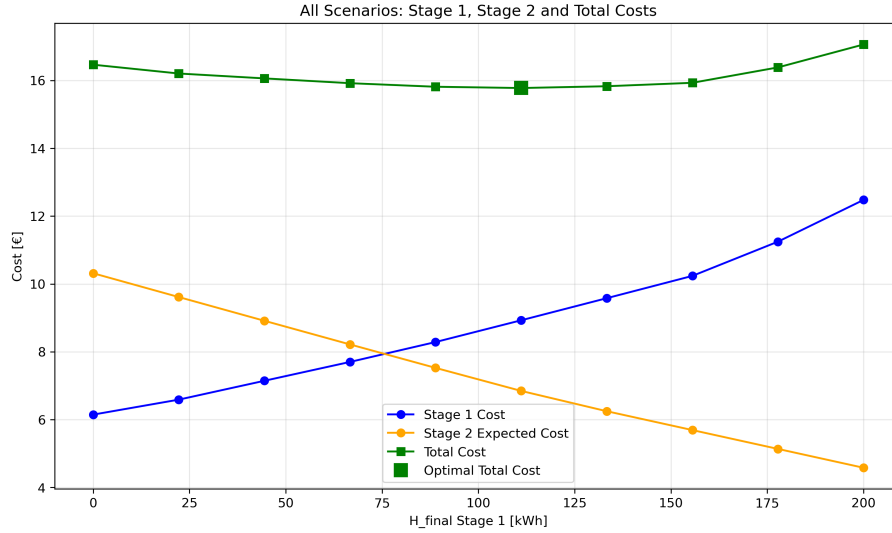


Figure 9: Stage 1 cost, Stage 2 expected cost, and total objective function as functions of hydrogen storage across all three scenarios.

The resulting optimal dispatch is qualitatively similar to Benders' solution, though the 4.7 kWh difference in optimal hydrogen (111.11 vs. 106.45 kWh) reflects SDP's discrete grid constraint. Since the discrete grid cannot exactly represent 106.45 kWh, SDP selects the nearest feasible point, leading to a slightly different Stage 2 cost profile.

Comparison with Benders Decomposition

Benders decomposition and SDP represent complementary approaches: Benders converges with exceptional precision to the exact optimum, while SDP characterizes the objective function across the entire state space.

Method	H_{23}^* [kWh]	Cost [€]	Time [s]
Benders Decomposition	106.45	15.7738	0.50
SDP (10 points)	111.11	15.7755	1.25

Table 9: Comparison Benders vs SDP.

The 4.7 kWh difference in H_{23} reflects these two fundamental distinctions. Benders identifies 106.45 kWh with high precision through iterative refinement near the optimum (0.50 s). SDP's discrete grid (spacing 22.22 kWh) cannot represent this exact value and selects 111.11 kWh instead. Both achieve nearly identical costs (0.17% difference), validating both approaches..

The first key difference is precision at optimality: Benders' cutting planes converge to the true cost-minimizing point with exceptional accuracy, making it ideal when cost minimization is the primary objective.

The second key difference is global landscape visibility: SDP uniformly evaluates all candidate hydrogen levels, explicitly characterizing how system costs vary across the entire state space. Benders, by contrast, concentrates refinement near optimality with limited characterization elsewhere. This comprehensive visibility makes SDP invaluable for sensitivity analysis and multi-stage extensions.

Figure 10 illustrates this trade-off: Benders shows concentrated refinement near $H_{23}^* = 106.45$ kWh with sparse approximation elsewhere; SDP displays uniform evaluation points revealing the full problem structure.

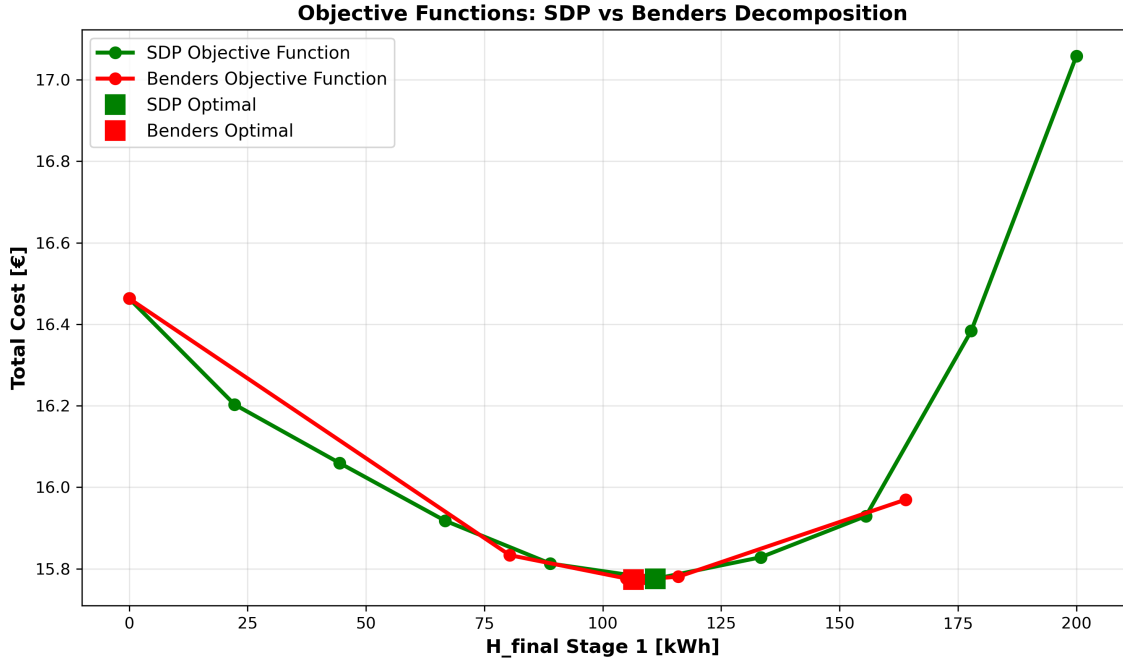


Figure 10: Objective function as a function of the linking variable (Benders vs SDP).

Comparison with Deterministic Equivalent

Both the deterministic equivalent (DE) and SDP solve the two-stage stochastic problem but employ fundamentally different strategies.

Method	H_{23}^* [kWh]	Cost [€]	Time [s]
Deterministic Equivalent	106.45	15.7738	0.20
SDP (10 points)	111.11	15.7755	1.25

Table 10: Comparison DE vs SDP.

The deterministic equivalent provides an exact solution with simple implementation and remarkable computational efficiency (0.20 s), making it particularly attractive for small-scale problems. Its primary limitation is that model size grows rapidly with the number of scenarios and stages, rendering it impractical for large-scale extensions.

Stochastic dynamic programming, by contrast, offers a naturally flexible framework for multi-stage decomposition with full characterization of the value function across the state space. The discretization introduces a 4.7 kWh deviation in optimal hydrogen storage and requires increased computational effort (1.25 s), but this overhead is offset by the method’s inherent ability to handle longer planning horizons and scenario aggregation without proportional problem growth.

Both methods achieve nearly identical total costs (0.17% difference), providing strong validation of the solution quality. The choice between them depends on problem scale and extension potential: DE excels for small-scale problems with a fixed number of scenarios and stages, while SDP becomes superior for multi-stage extensions (3+ stages) where its decomposition advantages become decisive and time savings are amplified.

Method Selection

Based on the results obtained from Parts 2–4, the choice of method depends on problem size, desired insights, and computational resources:

- **Expected Value (EV) Approach:** Simplest approach; solves one deterministic problem based on average inflows. Does not consider variability. It leads to a total cost of **15.68 €**
- **Deterministic Equivalent (DE):** Provides exact optimal solution under uncertainty with non-anticipativity constraints. Computationally efficient for small number of scenarios, with total cost **15.77 €** and hydrogen $H_{23}^* = 106.45$ kWh.
- **Benders Decomposition:** Efficient for larger problems; reproduces DE results while splitting the problem into master and subproblems. Ideal when scenario count is high or model is large-scale.
- **Stochastic Dynamic Programming (SDP):** Offers a flexible, state-based decomposition. Captures Stage 1–Stage 2 trade-offs and allows scenario aggregation. Slightly higher computational time (1.25 s), optimal hydrogen $H_{23}^* = 111.11$ kWh, total cost **15.78 €**. Particularly useful for multi-stage extensions beyond two stages.

In conclusion, for the current 48-hour, 3-scenario problem, the deterministic equivalent or Benders decomposition is preferred due to simplicity and speed. SDP provides comparable results and additional insights, making it attractive for larger or more complex multi-stage stochastic problems.

Appendix

Part 2 – Expected Value Approach

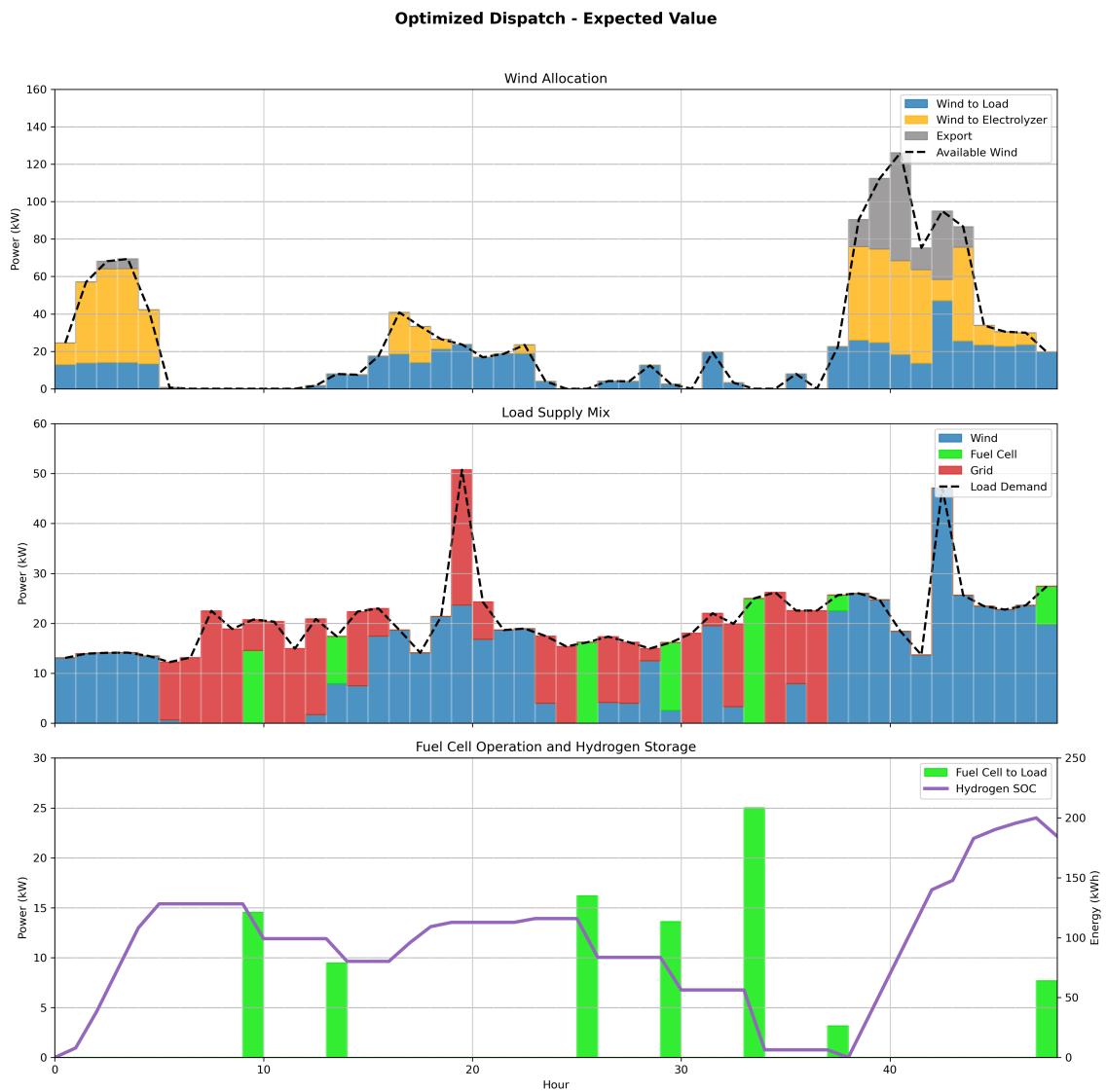


Figure 11: Hourly power dispatch across wind to load, electrolyzer, and exports under the Expected Value approach.

Part 2 – Deterministic Optimization

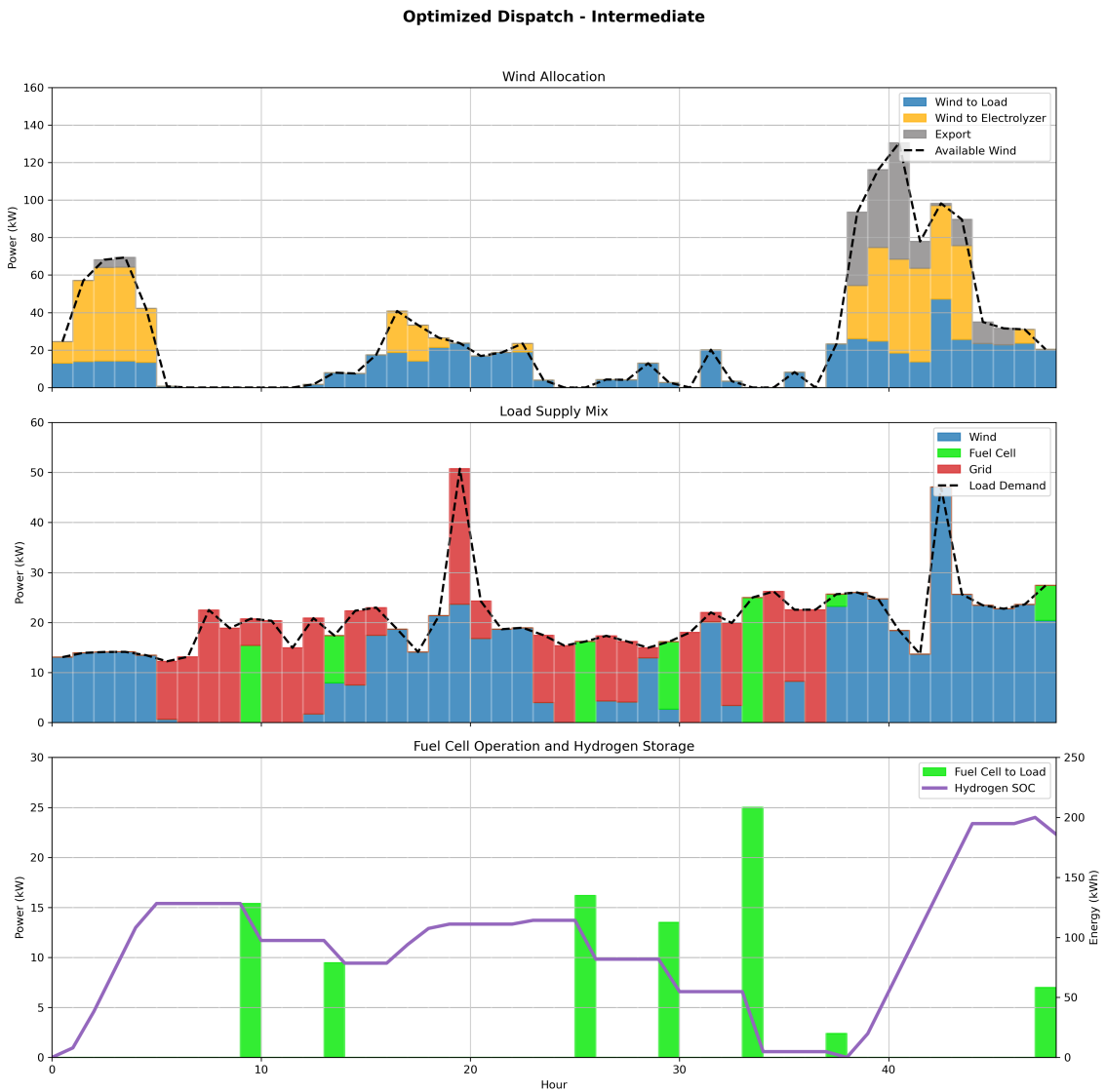


Figure 12: Hourly power dispatch across wind to load, electrolyzer, and exports for the intermediate scenario (deterministic).

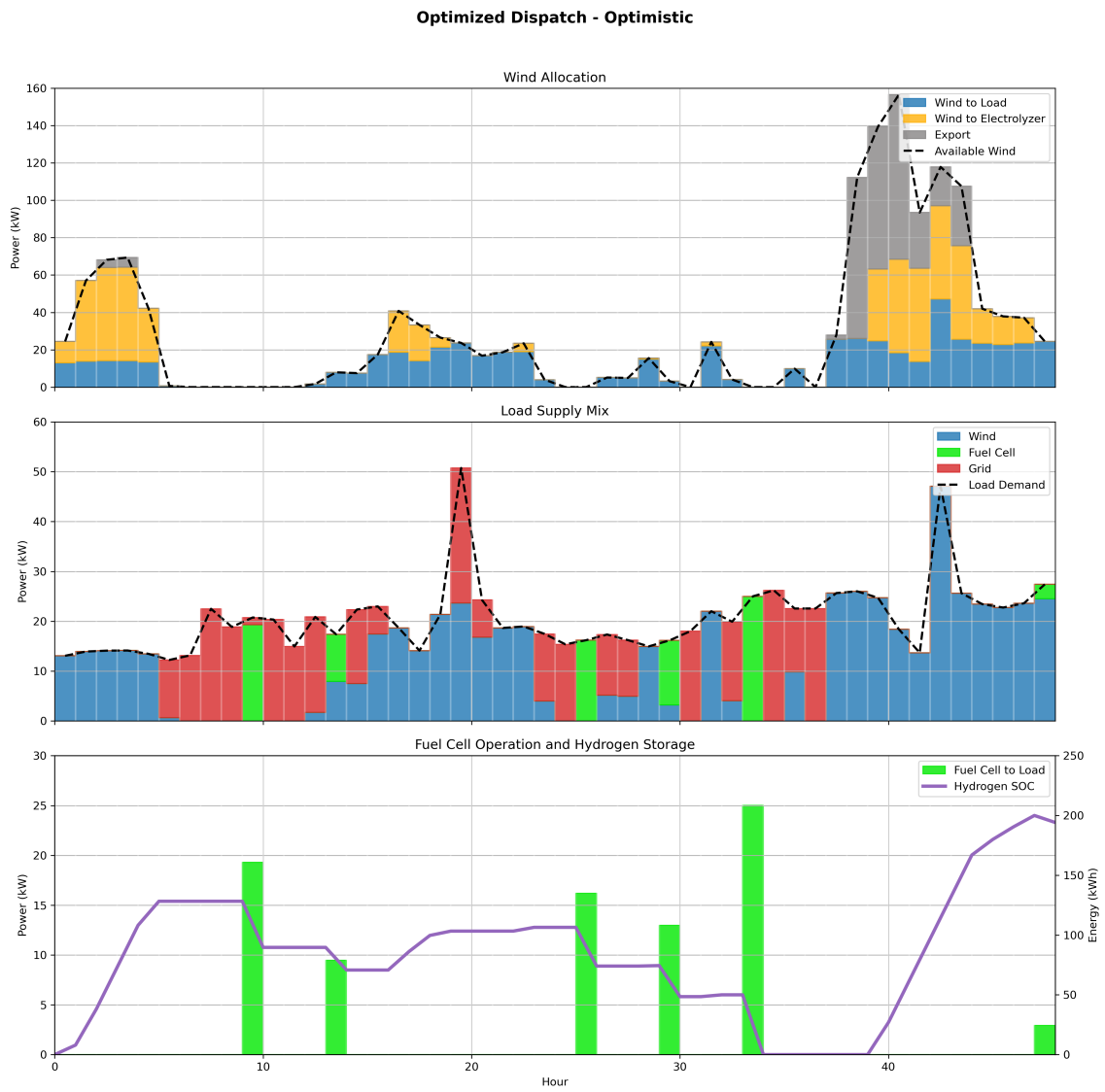


Figure 13: Hourly power dispatch across wind to load, electrolyzer, and exports for the optimistic scenario (deterministic).

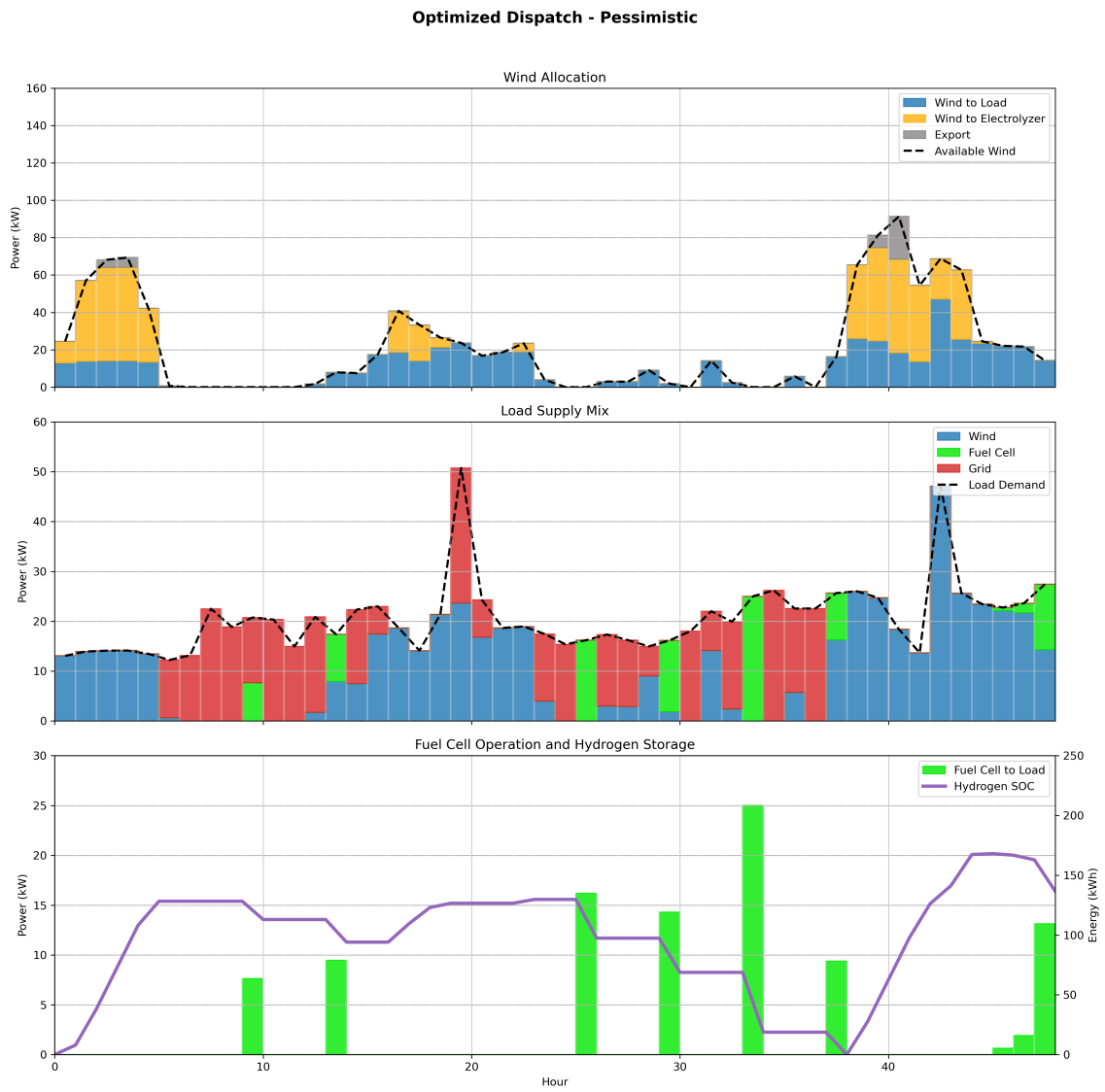


Figure 14: Hourly power dispatch across wind to load, electrolyzer, and exports for the pessimistic scenario (deterministic).

Part 2 – Stochastic Optimization (Additional Scenarios)

The following figures present the stochastic optimization results for the optimistic and pessimistic scenarios for completeness.

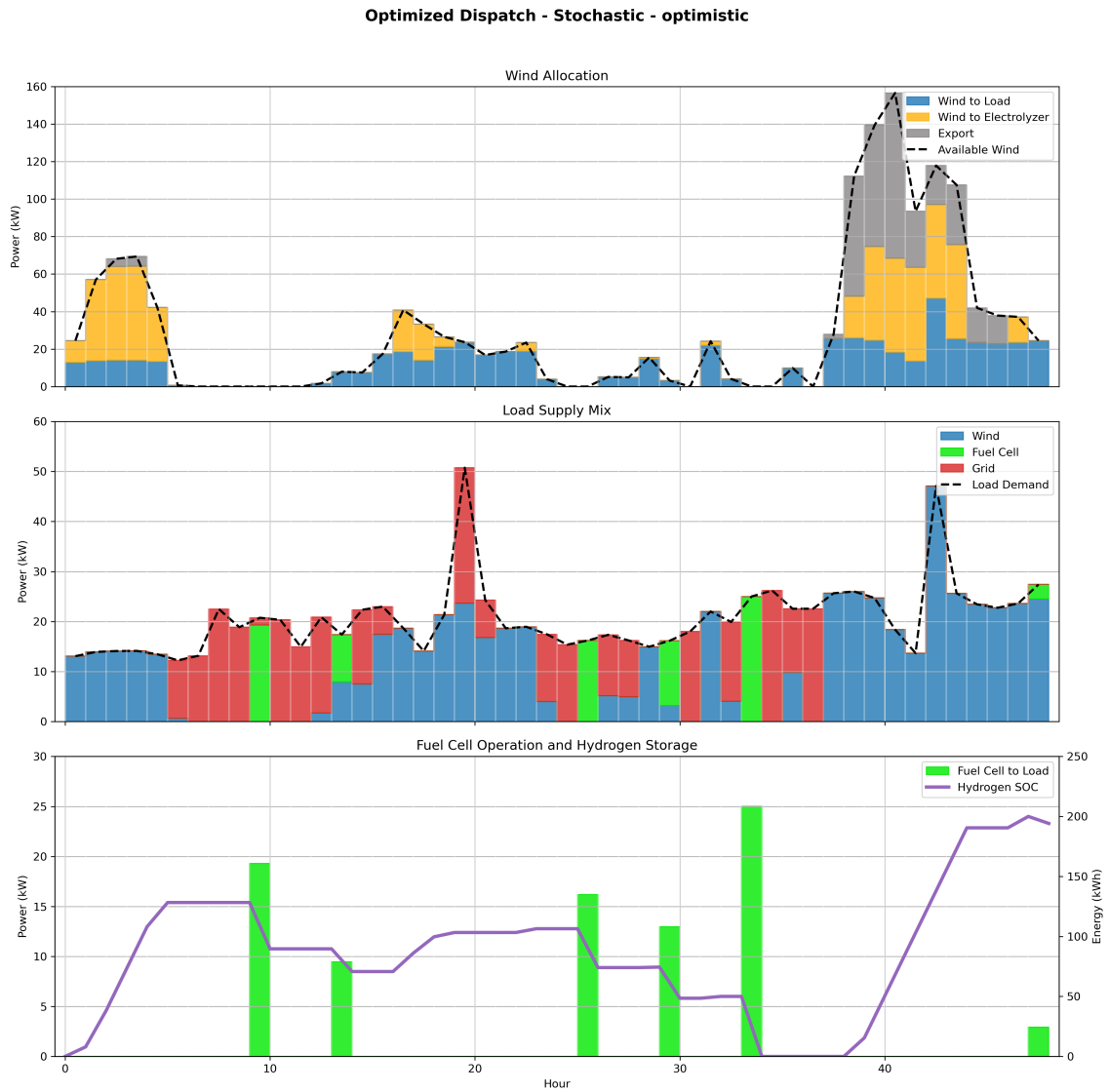


Figure 15: Hourly power dispatch across wind to load, electrolyzer, and exports under the stochastic approach for the optimistic scenario.

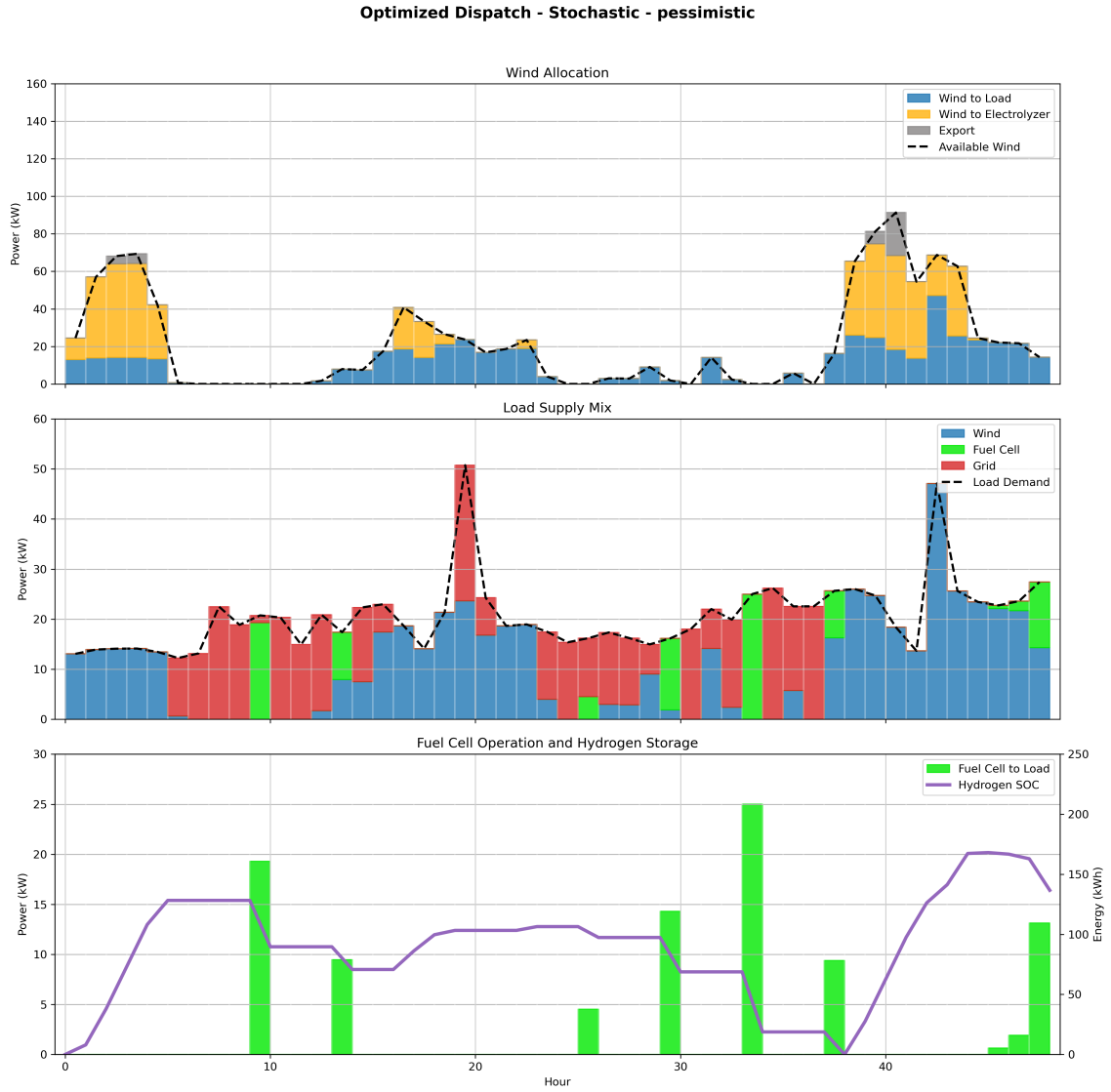


Figure 16: Hourly power dispatch across wind to load, electrolyzer, and exports under the stochastic approach for the pessimistic scenario.

Note: Benders decomposition results for the optimistic and pessimistic scenarios are omitted as they are identical to the corresponding stochastic optimization results presented above.